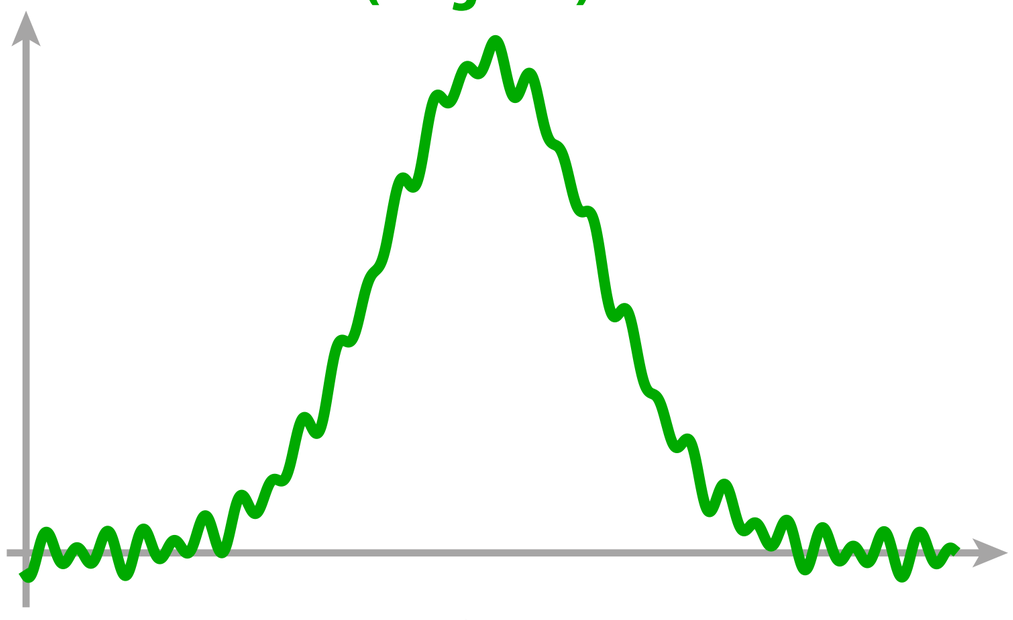
Aliasing

Sampling & Reconstruction

continuous signal (original)



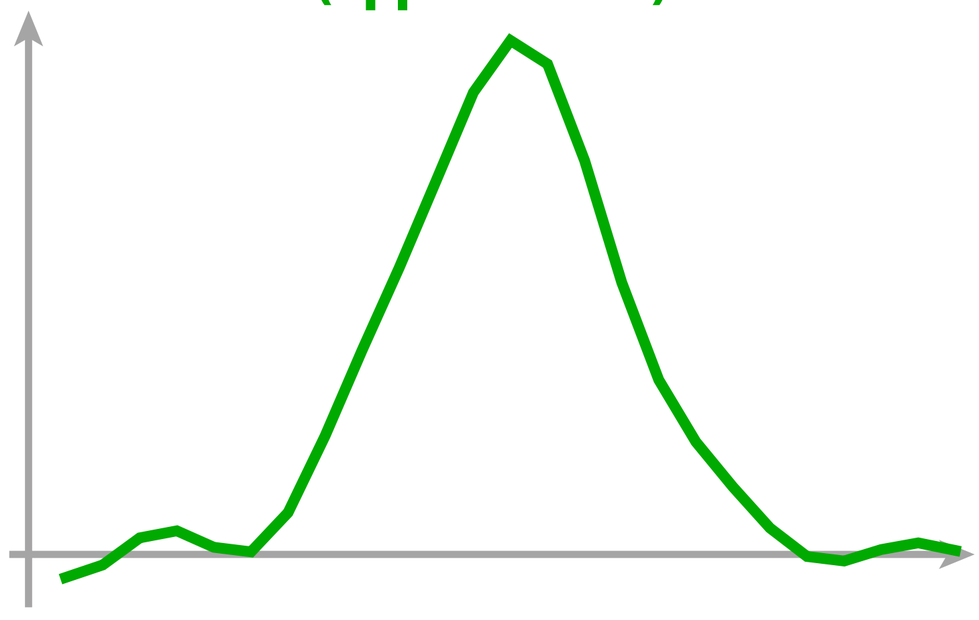
sample

```
-0.0457447
-0.0209434
 0.0328632
 0.0468068
 0.0141212
0.00506562
 0.0829571
  0.235369
  0.405682
  0.569204
  0.742511
  0.916946
      1.02
  0.973226
  0.781528
  0.539974
    0.3464
  0.223501
  0.134011
 0.0523279
-0.00416744
-0.0131345
 0.00959633
 0.0228676
 0.00789505
```

digital information

reconstruct

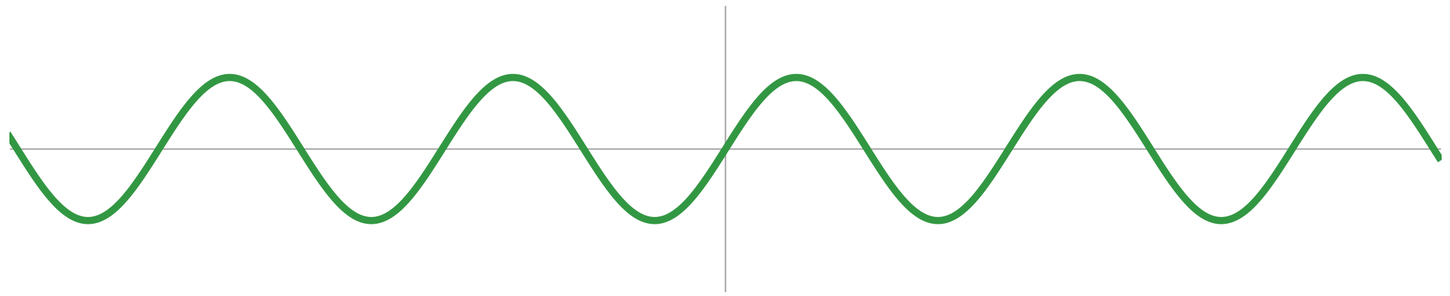
continuous signal (approximate)



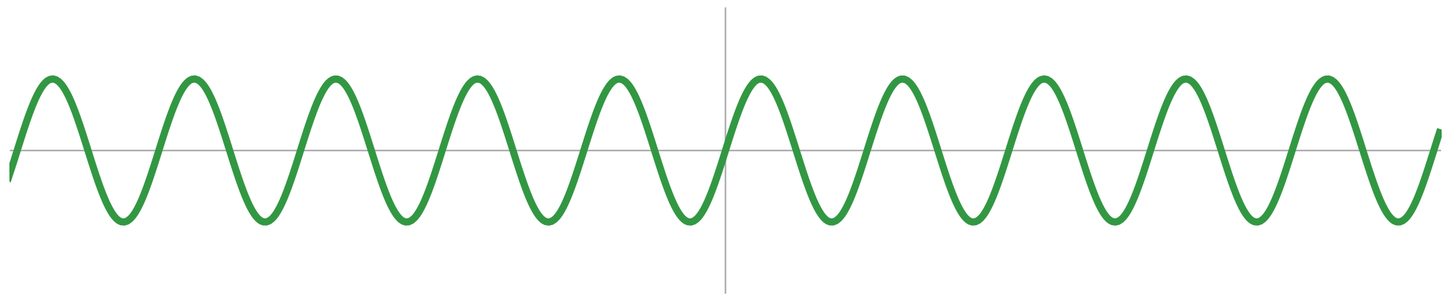
Goal: reproduce original signal as accurately as possible.

1D signal can be expressed as a superposition of frequencies

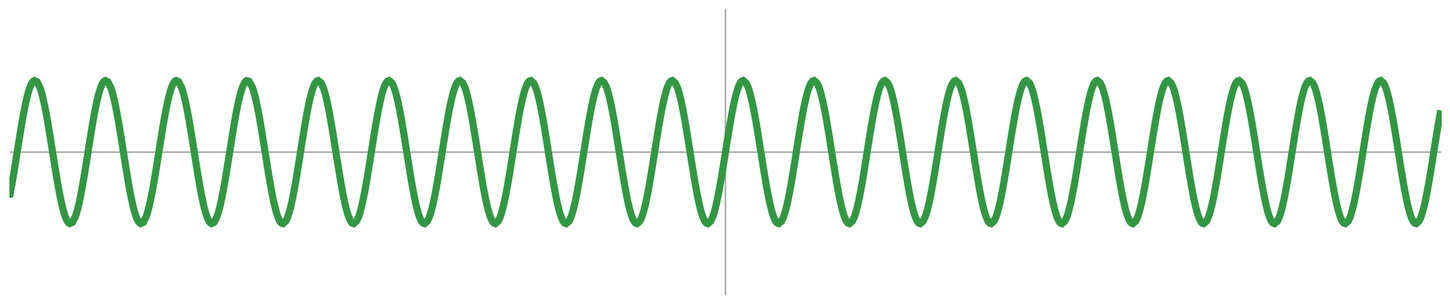
$f_1(x) = sin(\pi x)$



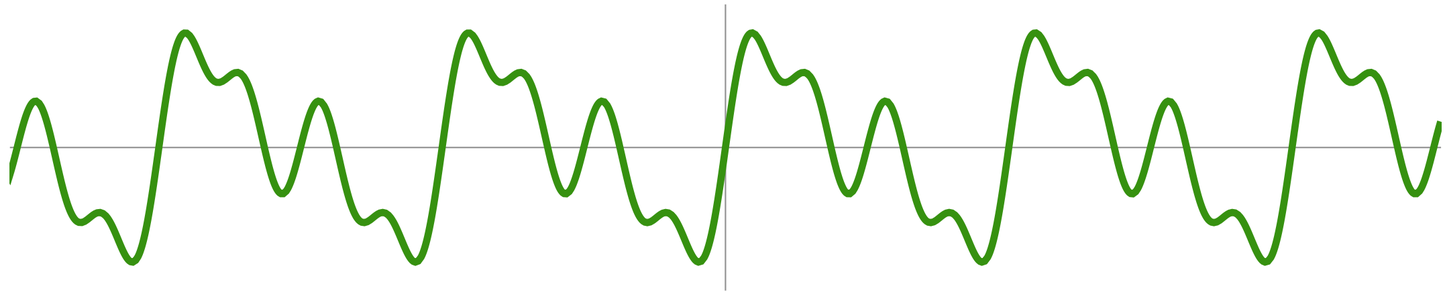
$f_2(x) = sin(2\pi x)$



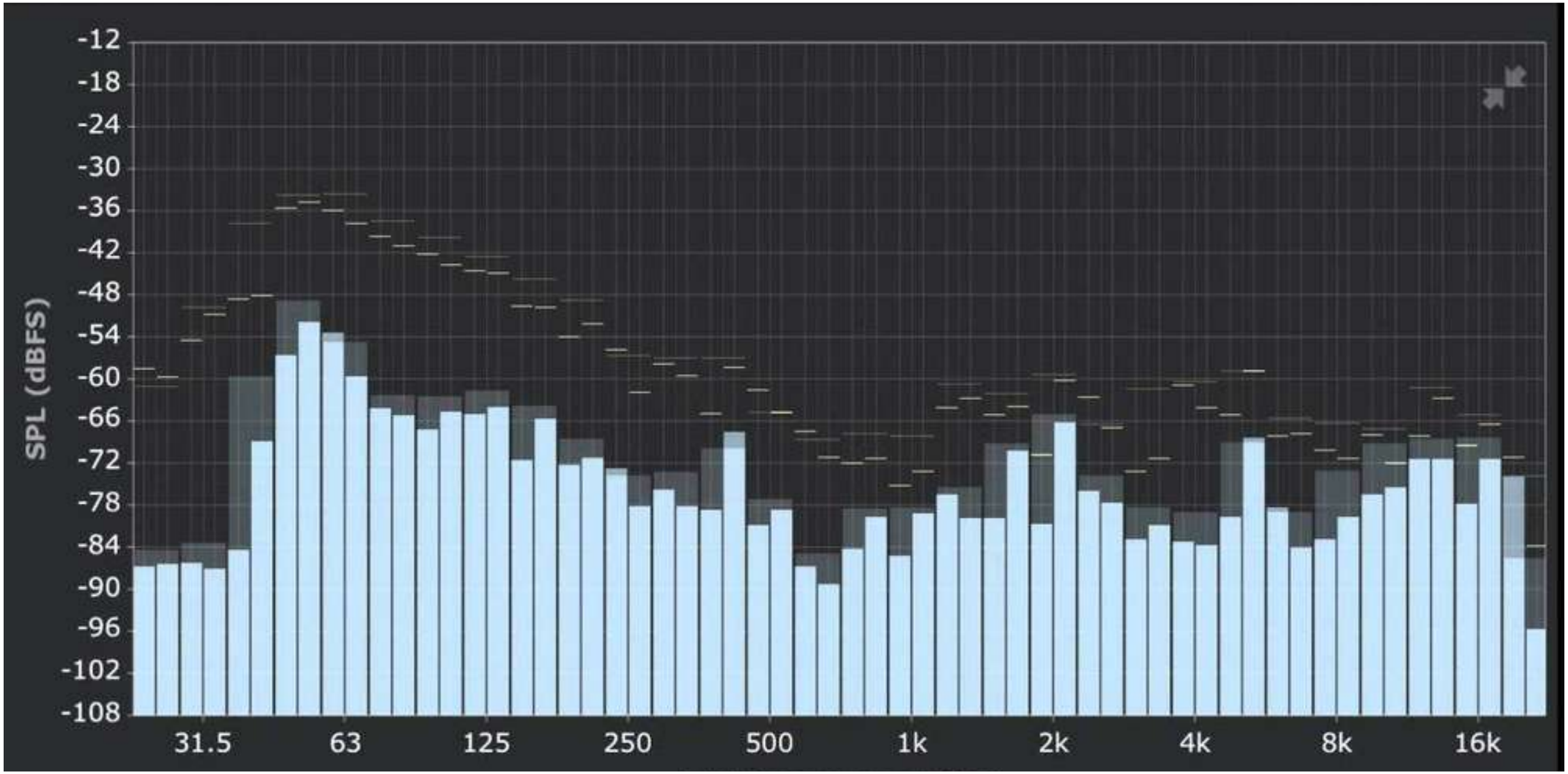
$f_4(x) = sin(4\pi x)$



$f(x) = f_1(x) + 0.75\, f_2(x) + 0.5\, f_4(x)$



E.g., audio spectrum analyzer shows the amplitude of each frequency



Intensity of low-frequencies (bass)

Intensity of high frequencies

Image credit: ONYX Apps

Aliasing in Audio

Get a constant tone by playing a sinusoid of frequency ω:

```
Play[Sin[4000 t], {t, 0, 1}]
```




```
Play[Sin[5000 t], {t, 0, 1}]
```

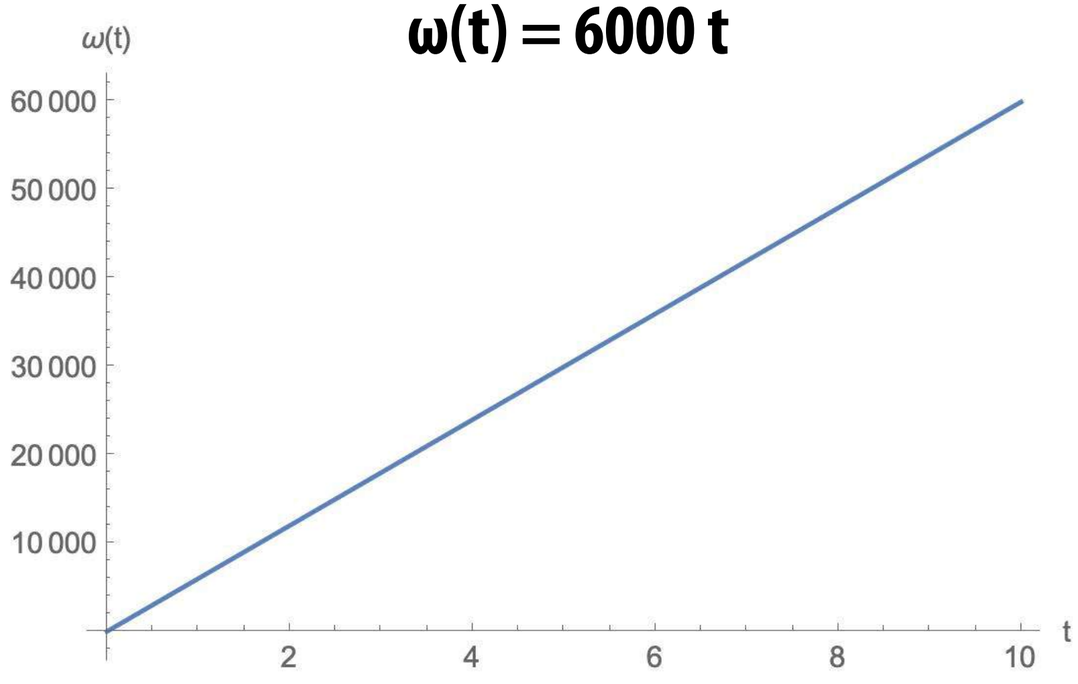



```
Play[Sin[6000 t], {t, 0, 1}]
```

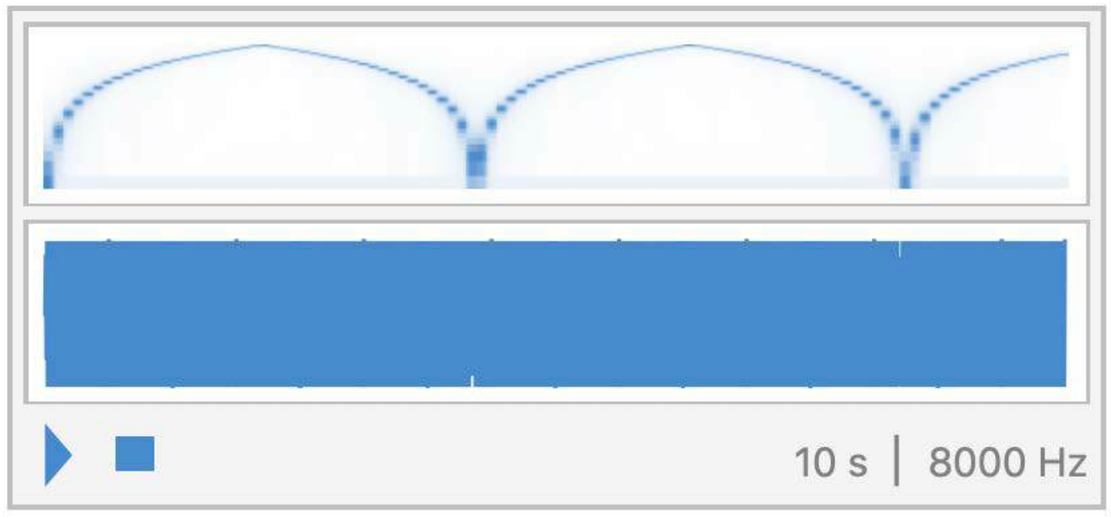


Q: What happens if we <u>increase</u> ω over time?

$\omega(t) = 6000\,t$

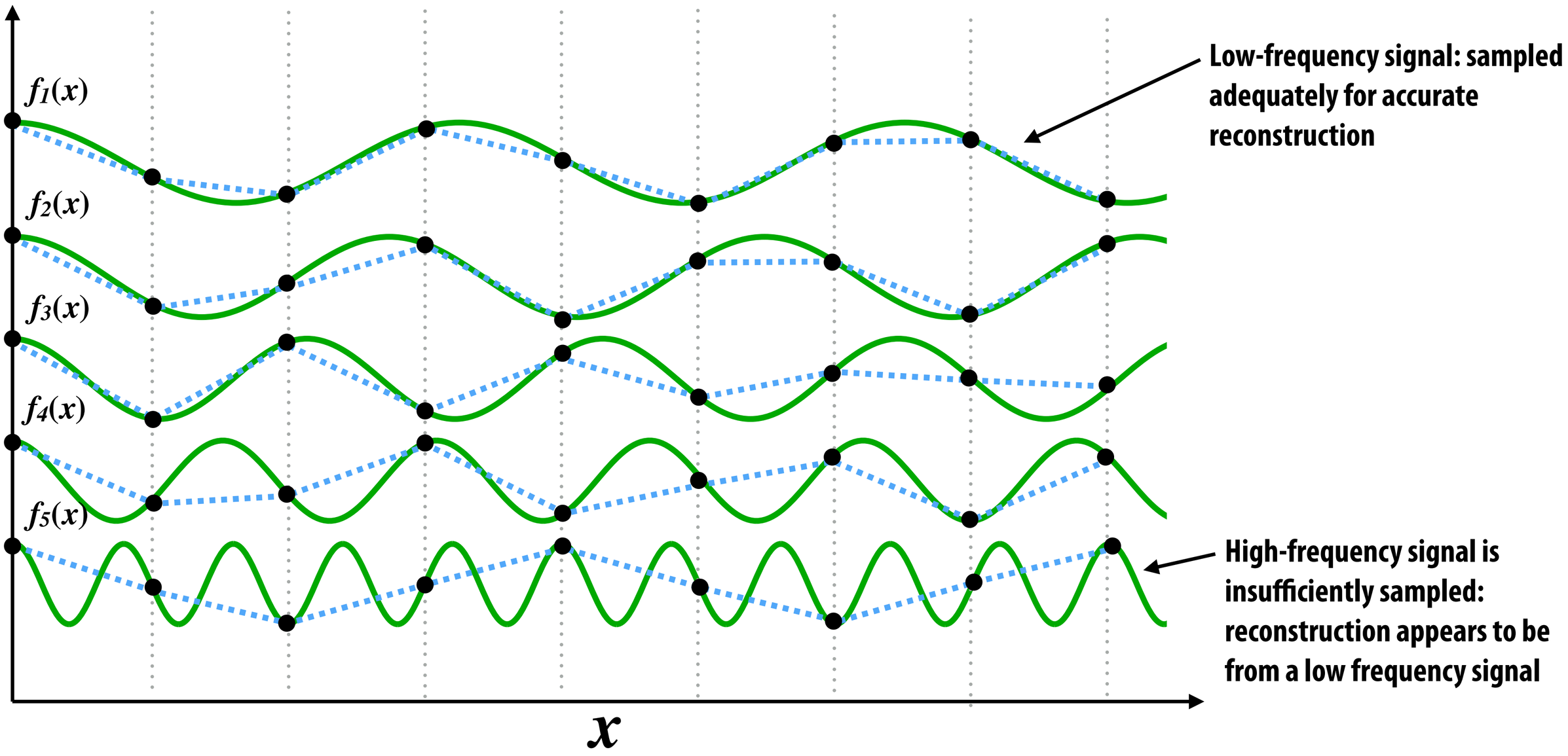



```
Play[Sin[ω t], {t, 0, 10}]
```



Why did that happen?

Undersampling high-frequency signals results in aliasing

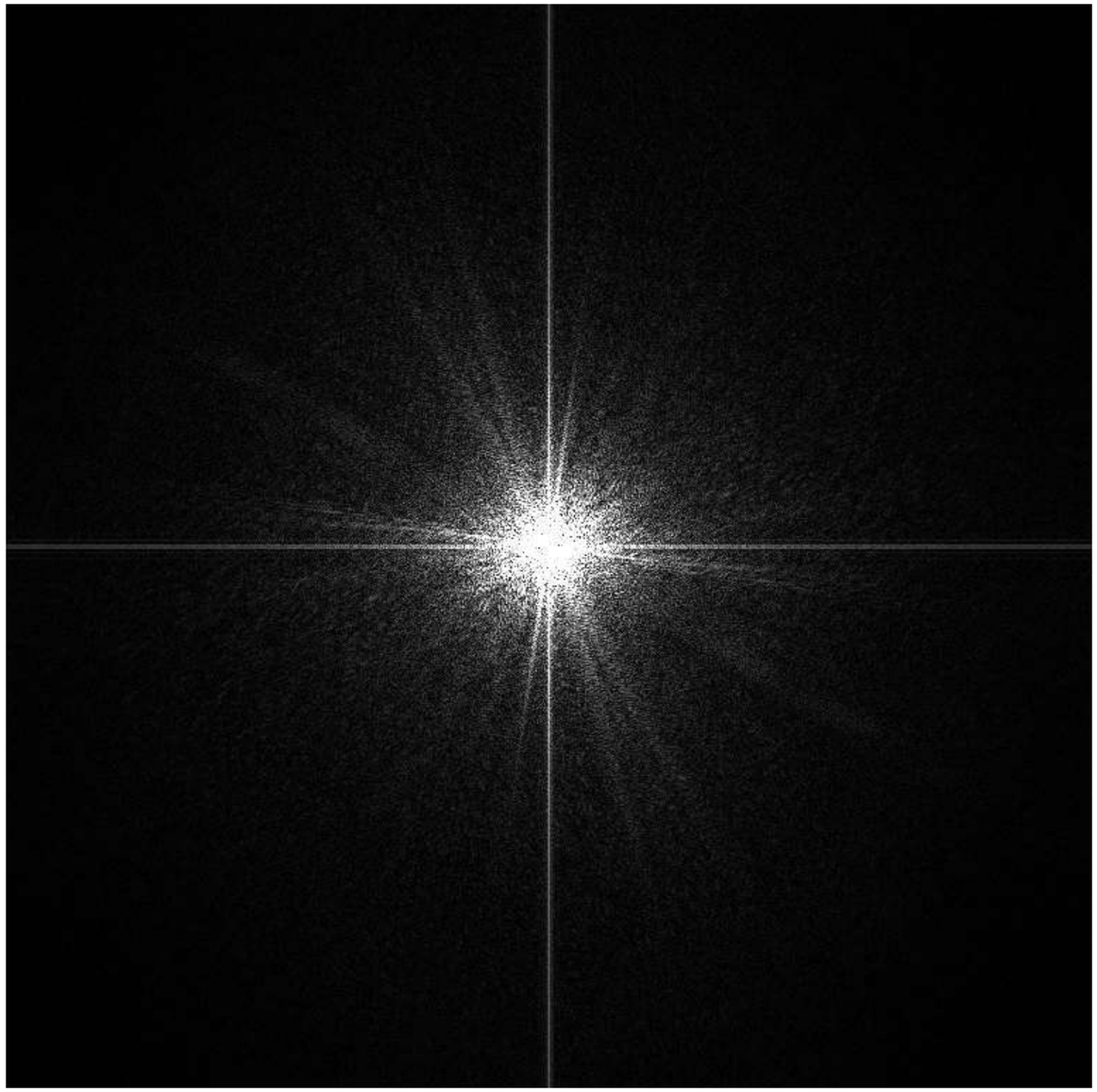


"Aliasing": high frequencies in the original signal masquerade as low frequencies after reconstruction (due to undersampling)

Images can also be decomposed into “frequencies”



Spatial domain result

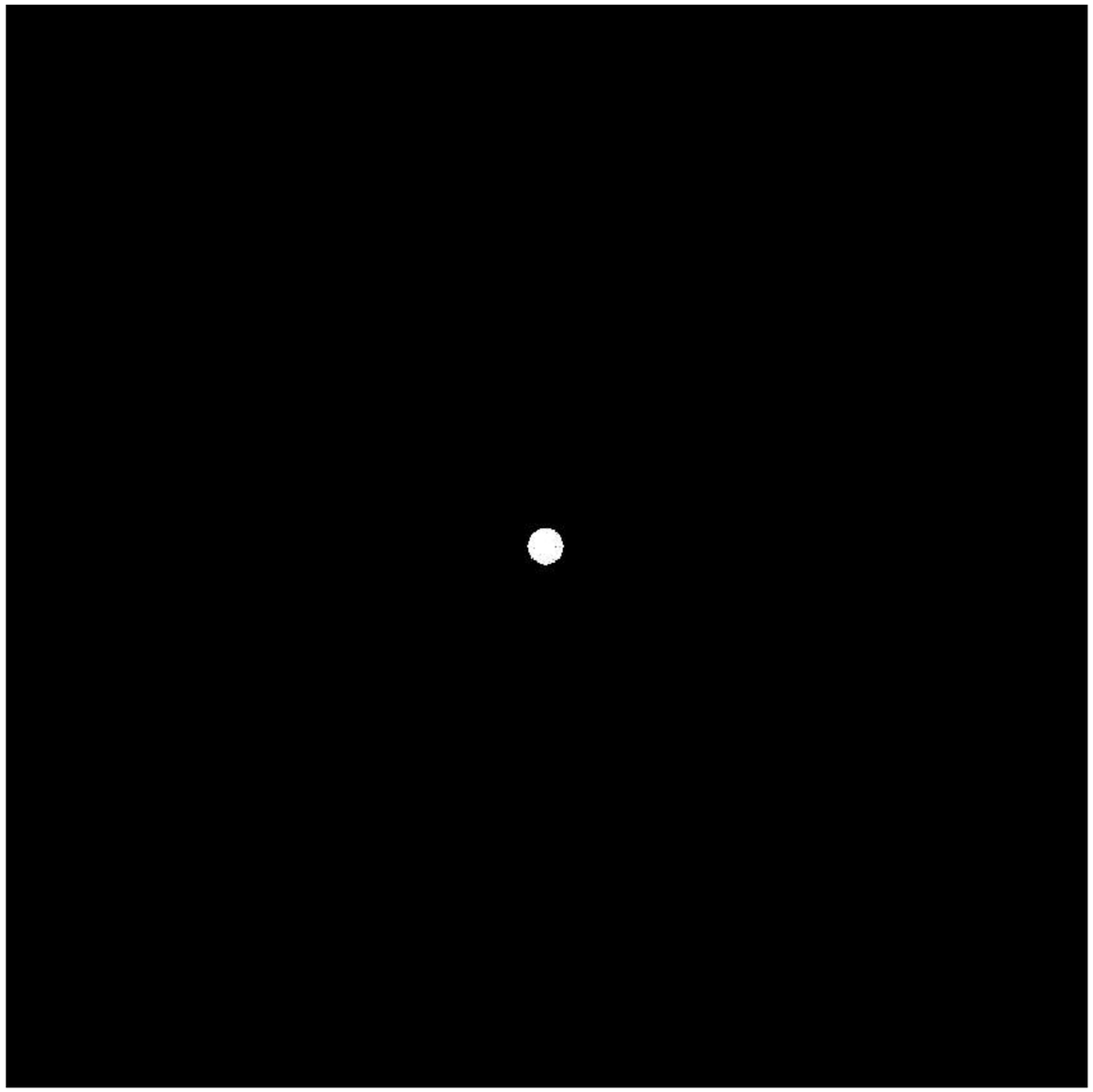


Spectrum

Low frequencies only (smooth gradients)



Spatial domain result

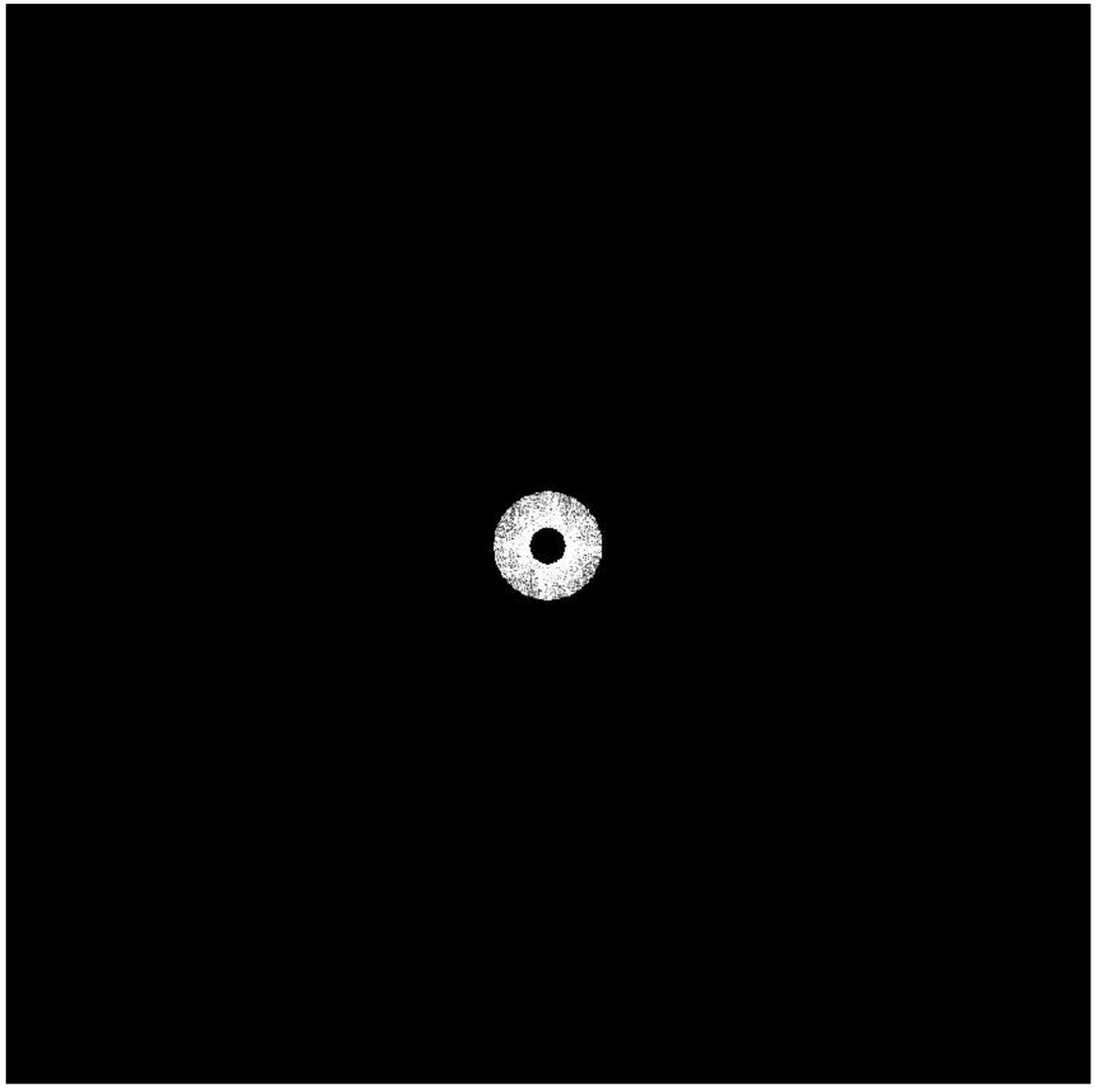


Spectrum (after low-pass filter)
All frequencies above cutoff have 0 magnitude

Mid-range frequencies



Spatial domain result

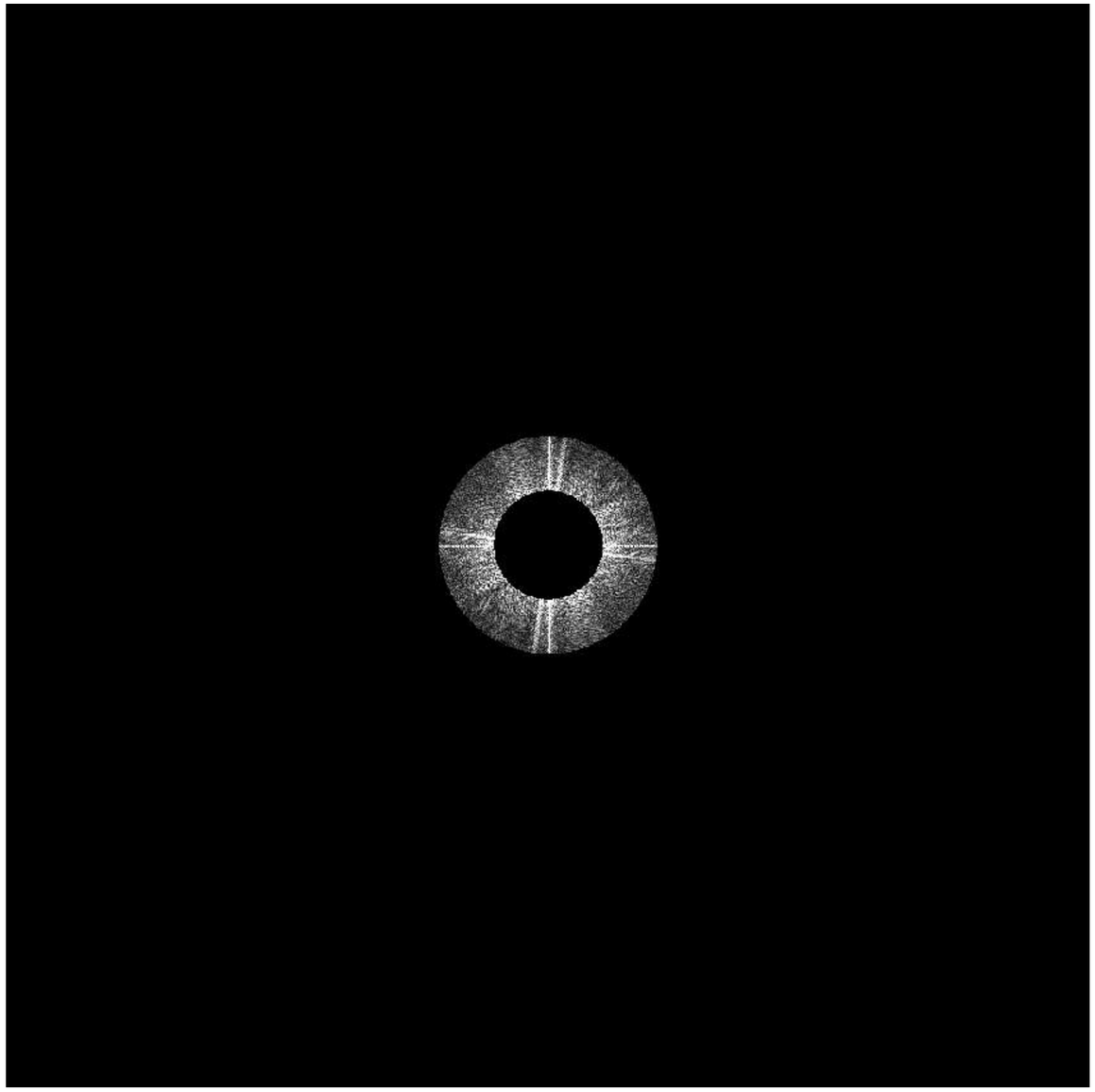


Spectrum (after band-pass filter)

Mid-range frequencies

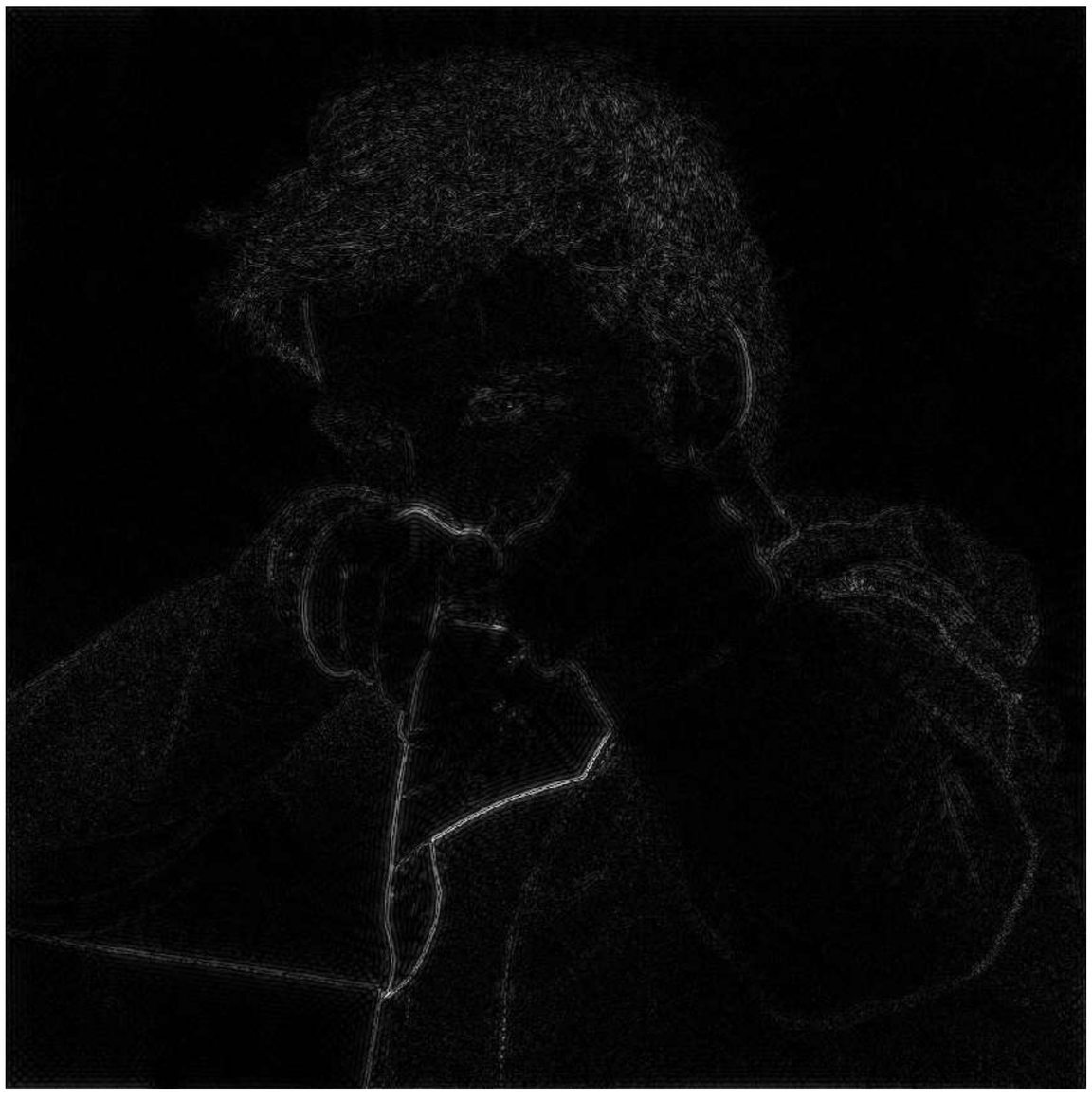


Spatial domain result

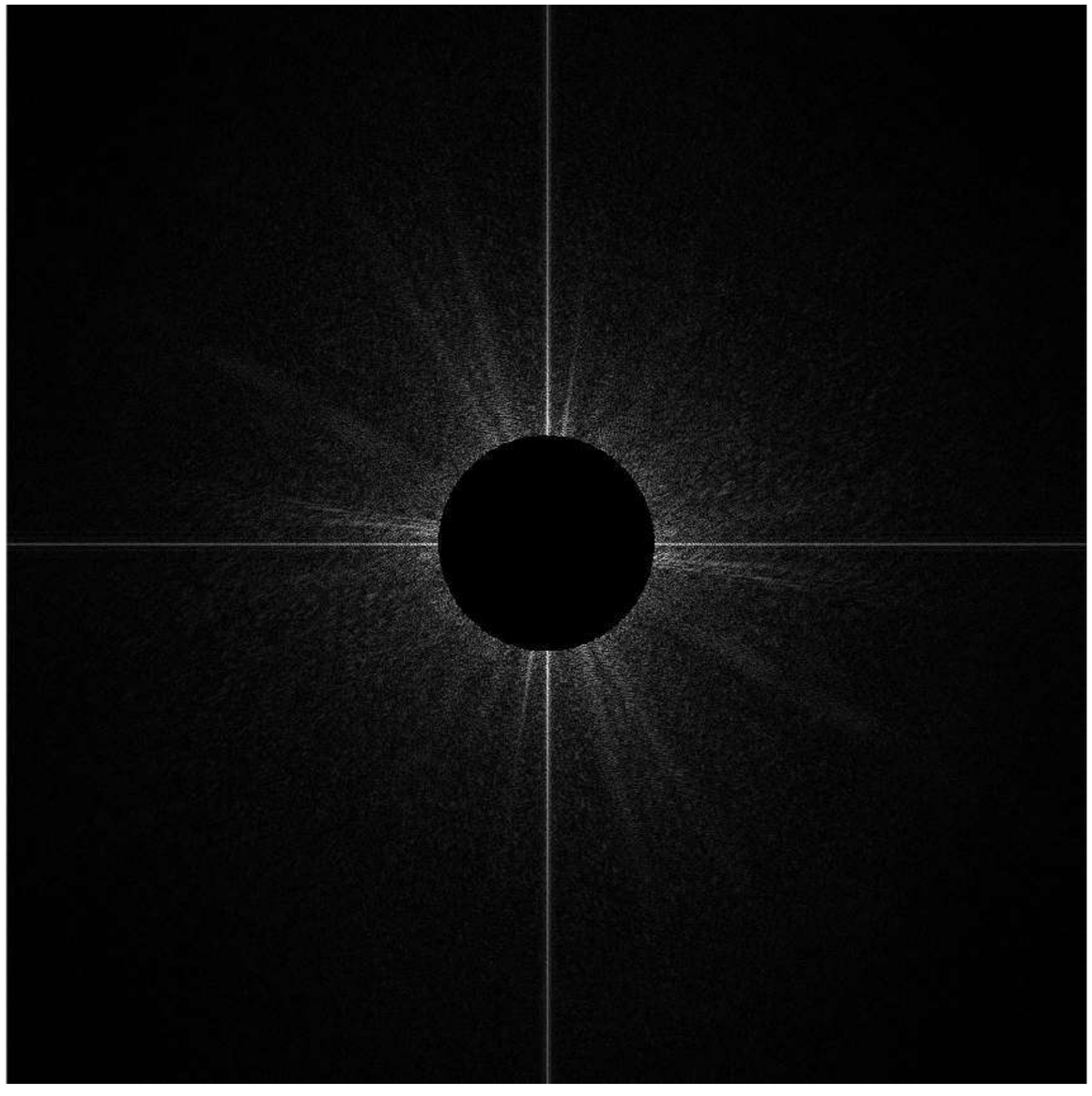


Spectrum (after band-pass filter)

High frequencies (edges)

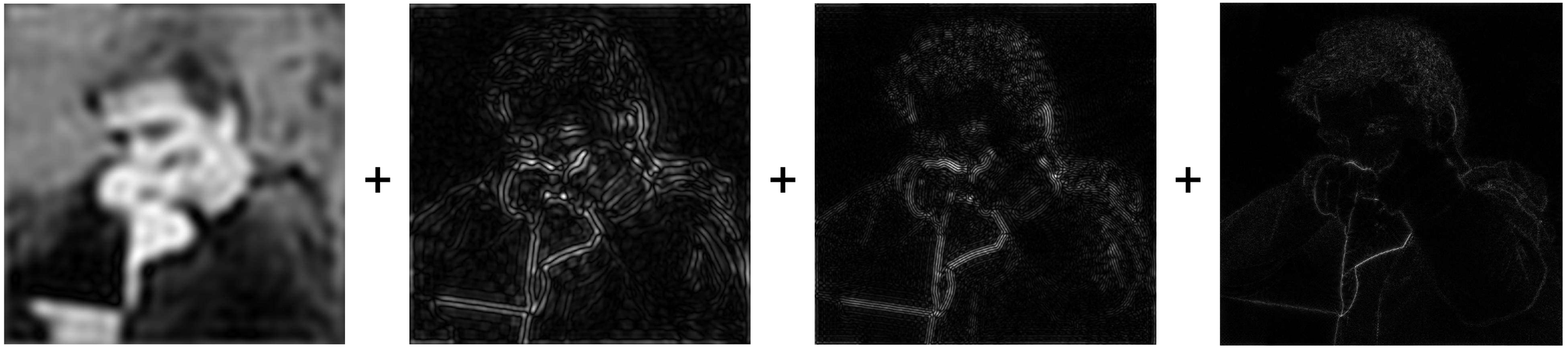


Spatial domain result
(strongest edges)



Spectrum (after high-pass filter)
All frequencies below threshold
have 0 magnitude

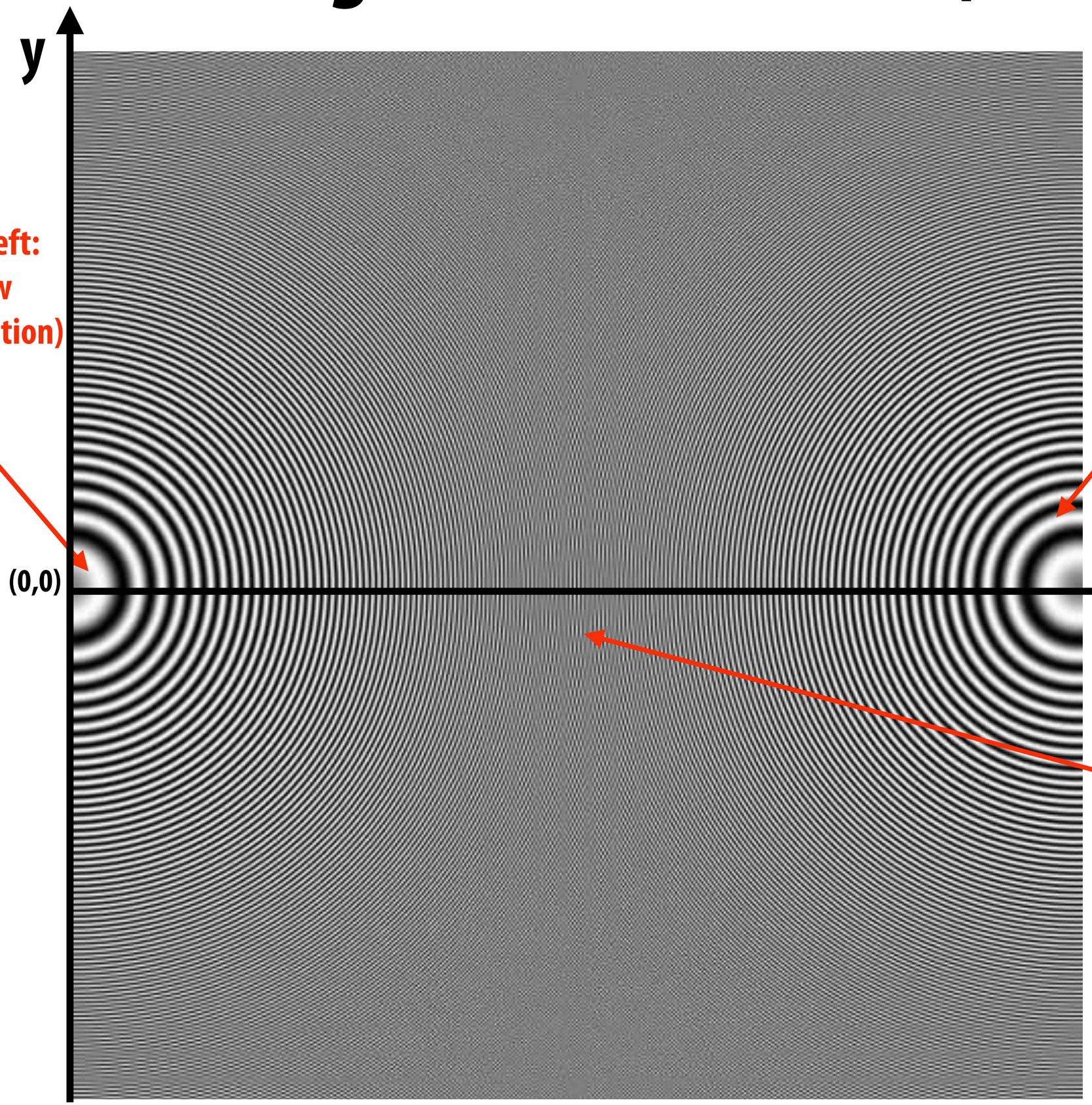
An image as a sum of its frequency components



=



Spatial aliasing: the function $\sin(x^2 + y^2)$



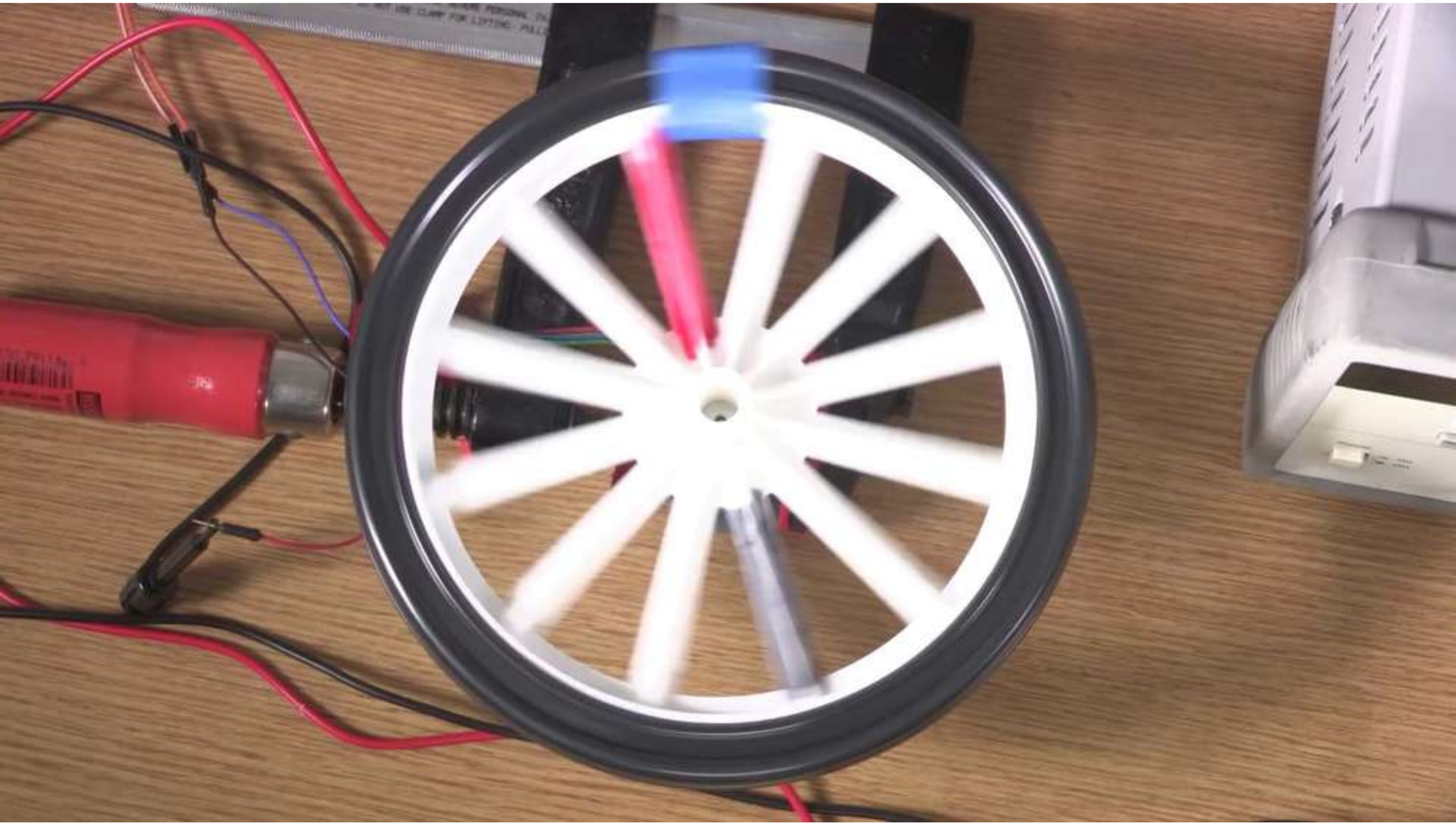
Rings in center-left: Actual signal (low frequency oscillation)

Right: aliasing from undersampling high frequency oscillation makes it appear that rings are low-frequency (they're not!)

Middle: ring frequency approaches limit of what we can represent w/ individual pixels

Figure credit: Pat Hanrahan and Bryce Summers

Temporal aliasing: wagon wheel effect



Camera's frame rate (temporal sampling rate) is too low for rapidly spinning wheel.